CSE422: Informed Search: Heuristic Analysis Problem Set

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Problem 1: Consistency of Heuristics in the 8-Puzzle

Scenario: You are solving the 8-puzzle on a 3×3 board using A*. Each move (sliding one tile into the blank) costs 1.

$$h_1(n) = \text{number of misplaced tiles}, \qquad h_2(n) = \sum_{i=1}^{8} (|x_i - x_i^G| + |y_i - y_i^G|).$$

Tasks

(a) Explain what each heuristic measures and why h_2 is more informative than h_1.

(b) Show that both h_1 and h_2 are **admissible**.

(c) Prove that both are **consistent**, i.e.

$$h(n) \leq 1 + h(n')$$

for any neighbor n' reached by one legal move.

(d) Show that h_2 **dominates** h_1.

Hints from Instructor:

- A misplaced tile must move at least once $\Rightarrow h_1$ never overestimates.
- Each tile's Manhattan distance counts minimal required steps $\Rightarrow h_2$ never overestimates.
- Moving one tile changes h_1 or h_2 by at most 1 $\Rightarrow$ both are consistent.
- For dominance, show $|x_i - x_i^G| + |y_i - y_i^G| \geq 1$ for each misplaced tile $\Rightarrow h_2 \geq h_1$.

Problem 2: Euclidean vs Manhattan on a 4-Connected Maze

Scenario: You are solving a maze on a 5×5 grid with Start $S(x_n, y_n)$ and Goal $G(x_G, y_G)$. Allowed moves: up, down, left, right (no diagonals). Each move costs 1.

$$h_E(n) = \sqrt{(x_n - x_G)^2 + (y_n - y_G)^2} = \sqrt{dx^2 + dy^2}, \qquad h_M(n) = |x_n - x_G| + |y_n - y_G| = dx + dy.$$

Tasks

(a) Prove h_E is **admissible** and **consistent**.

(b) Prove h_M is **admissible** and **consistent**.

(c) Determine which heuristic **dominates** the other and justify.

Hints from Instructor:

- For admissibility: The true path cost $\geq dx + dy$ steps (Manhattan distance) $\geq$ straight-line Euclidean distance. Hence both are lower bounds.
- For consistency: A single 4-connected move changes either dx or dy by ± 1. Show $h(n) - h(n') \leq 1 = c(n, n')$.
- For dominance: Since $(dx+dy)^2 = dx^2+dy^2+2dx\,dy \geq dx^2+dy^2$, $h_M(n) = dx+dy \geq \sqrt{dx^2 + dy^2} = h_E(n)$, so Manhattan dominates Euclidean on a 4-connected grid.

Problem 3: Consistent Heuristic for Terrain-Dependent Maze

Scenario: You are solving a 4-connected maze (5×5 grid). Each move (u, v) has a terrain-dependent cost $c(u, v) \geq c_{\min} > 0$. Goal position: $G(x_G, y_G)$. For node $n(x_n, y_n)$ define:

$$dx = |x_n - x_G|, \quad dy = |y_n - y_G|.$$

Tasks

(a) Propose a heuristic $h(n)$ that is both **admissible** and **consistent** under varying costs.

(b) Show its admissibility.

(c) Show its consistency.

(d) Compare two scaled heuristics:

$$h_M(n) = c_{\min}(dx + dy), \quad h_E(n) = c_{\min}\sqrt{dx^2 + dy^2},$$

and determine which dominates.

Hints from Instructor:

- Any valid path must include at least dx horizontal and dy vertical moves, each costing at least $c_{\min}$. Thus $c_{\min}(dx + dy)$ is a valid lower bound.
- A scaled Euclidean form $c_{\min}\sqrt{dx^2 + dy^2}$ is also admissible.
- For consistency, check that moving one step changes $dx + dy$ by at most 1, so $h(n) - h(n') \leq c_{\min} \leq c(n, n')$.
- For dominance: On 4-connected grids, $dx + dy \geq \sqrt{dx^2 + dy^2}$, so h_M dominates h_E.